

Data cleaning, transformation and modelling

Q1. What is Data Loading in Power BI and why is it considered the first step of analysis?

Ans. Data Loading in Power BI is the process of importing data from different sources such as Excel files, databases, websites, or cloud services into Power BI. It is considered the first step of analysis because before creating charts, reports, or dashboards, the data must first be collected and loaded into the software. Proper data loading helps in accurate analysis and better decision-making.

Data loading is considered the foundational first step of analysis for several critical reasons:

- **Foundation for All Metrics:** You cannot build visuals, write DAX measures, or create relationships without data. Every subsequent step relies entirely on the data being present.
- **Enables Data Transformation (ETL):** Loading data triggers Power Query, allowing you to clean, filter, reshape, and fix errors in your dataset before analysis begins.
- **Defines the Data Model:** The act of loading data is where you establish the tables and relationships that dictate how your data interacts, forming the bedrock of your entire report architecture.

Q2. Explain the difference between “Load” and “Transform Data” in Power BI.

Ans.

- **Load:**
The Load option directly imports the selected data into Power BI without making any changes. It is used when the data is already clean and ready for analysis.
- **Transform Data:**
The Transform Data option opens the Power Query Editor where users can clean and modify the data before loading it. For example, removing duplicate values, changing column names, filtering rows, or correcting data types.

Difference:

- Load imports data directly.
- Transform Data allows editing and cleaning the data before importing.

Q3. What is a Fact Table and a Dimension Table? Give examples from the dataset.

Ans. A Fact Table is a table that contains numerical values or measurable data used for analysis, such as sales amount, quantity, profit, or revenue. It usually includes foreign keys connected to dimension tables.

Example: A Sales Table containing Sales Amount, Quantity Sold, and Profit is a Fact Table.

A Dimension Table is a table that contains descriptive information related to the fact table, such as customer details, product names, dates, or locations. It helps in filtering and categorizing data.

Example: A Customer Table with Customer Name, City, and Region or a Product Table with Product Name and Category is a Dimension Table.

Q4. Why is Star Schema preferred over Snowflake Schema in Power BI?

Ans. Star Schema is preferred over Snowflake Schema in Power BI because it is simpler, faster, and easier to understand. In a Star Schema, the fact table is directly connected to dimension tables, which improves query performance and makes report creation easier.

Advantages of Star Schema:

1. Simple structure and easy to manage.
2. Faster data processing and better performance.
3. Easier to create relationships and visualizations in Power BI.
4. Reduces complexity for users and improves report readability.

In comparison, Snowflake Schema has multiple related tables, which makes the model more complex and can slow down performance.